

Medical Data as the Backbone of AI

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Duke Biostatistics & Bioinformatics

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Lecture Outline

- What types of data are in electronic health records (EHRs), and why?
 - The importance of considering **multiple data types**
 - Similarities and differences between EHRs and **other sources of medical data**
- Medical data as raw material for AI systems
 - **One** prediction task, **one** type of input data (task-**specific**, **uni**-modal)
 - **Any** prediction task, **one** type of input data (task-**agnostic**, **uni**-modal)
 - **One** prediction task, **multiple** types of input data (task-**specific**, **multi**-modal)
 - **Any** prediction task, **multiple** types of input data (task-**agnostic**, **multi**-modal)

The Electronic Health Record

One stop shop for most varieties of medical data

Structured Data

Definition: data organized into specific, queryable fields as part of an overarching model, with each field having a defined purpose.

- typically stored in tabular (i.e. table, spreadsheet) format
- numeric variables, categorical variables and/or short free text

Examples: demographics, diagnoses, procedures, labs, meds, vitals

An example of structured EHR data (PCORnet CDM)

PCOR_RSH_DIAGNOSIS_VW EXAMPLE

DIAGNOSISID	PATID	ENCOUNTERID	ENC_TYPE	ADMIT_DATE	PROVIDERID	DX	DX_TYPE	DX_SOURCE	DX_ORIGIN	PDX	DX_POA	RAW_DX	RAW_DX_TYPE	RAW_DX_SOURCE	RAW_PDX
36510180	123456	57896636	AV	2014/02/21 00:00:00	1745326	530.9	9	FI	BI	OT	NI	530.9	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY
36510407	223456	57897312	AV	2014/05/07 00:00:00	73411	530.81	9	FI	BI	OT	NI	530.81	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY
36510405	323456	57894371	ED	2014/06/13 00:00:00	123901	553.3	9	FI	BI	OT	NI	553.3	ICD-9-CM	FINAL DIAGNOSIS	SECONDARY
26510406	423456	57843289	AV	2015/01/21 00:00:00	1719263	530.3	9	FI	BI	OT	NI	530.3	ICD-9-CM	FINAL DIAGNOSIS	SECONDARY
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36508682	543216	57893648	IP	2016/08/05 00:00:00	178238	211.3	9	FI	BI	OT	NI	211.3	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY

COLUMN	DEFINITION
DIAGNOSISID:	An arbitrary identifier for each record. Not currently useful for linkage across tables.
ENCOUNTERID:	The ENCOUNTERID (encounter ID) is an arbitrary identifier for a specific encounter that can be used to link information across tables in the model.
PATID:	The PATID (patient identifier) is an arbitrary, person-level identifier used to link information across tables within the data model. There is a table within the CRDM that allows an investigator with an appropriate IRB protocol to link PATID to MRN.
ENC_TYPE:	Encounter type indicates the setting in which the encounter took place (the same data are provided in the encounter table: AV=Ambulatory Visit, ED=Emergency Department, EI=Emergency Department Admit to Inpatient Hospital Stay, IP=Inpatient Hospital Stay, OS=Observation Stay, NI=No information, UN=Unknown, OT=Other
ADMIT_DATE:	The date on which the encounter began (also included in the encounter table).
PROVIDERID:	Unique identifier for a provider ? can be used to link an encounter to the provider table.
DX:	The diagnosis code; uses either ICD-9 or ICD-10
DX_TYPE:	09=ICD-9-CM, 10=ICD-10-CM, 11=ICD-11-CM, SM=SNOMED CT, NI=No information, UN=Unknown, OT=Other

Unstructured Data (i.e., everything else)

Definition: data that is NOT structured data

- not natural to organize in tabular format
- typically stored in a modality-specific format

Examples:

- clinical notes
- imaging (e.g. radiology, ophthalmology, ultrasound, pathology)
- physiologic time series (e.g. electrophysiologic studies, digital health)

Is structured data enough?

What's contained in structured fields, and what's missing?

A patient presents with shortness of breath...

- Presents to urgent care and is transferred to the ED
- Evaluated in the ED
 - History of present illness
 - Review of medical history (including meds and allergies) + social and family hx
 - Physical exam and review of systems
 - Lab results: BMP and CBC
 - Imaging & other studies: CT and EKG
- Admitted to the hospital

What will we find in the structured data?

- Encounter details (ED admission)
 - Labs (BMP, CBC)
 - Vital signs
 - Procedure codes (CT)
 - Diagnosis codes (PE)
-
- ...what *won't* we find?

The H&P

written by a UNC student and publicly available:

<https://www.med.unc.edu/medclerk/education/grading/history-and-physical-examination-h-p-examples/>

FHIR systems **SHALL** support, at *minimum*, these eight “Common Clinical Notes”:

1. Consultation Note (11488-4)
2. Discharge Summary (18842-5)
3. History & Physical Note (34117-2)
4. Procedures Note (28570-0)
5. Progress Note (11506-3)
6. Imaging Narrative (18748-4)
7. Laboratory Report Narrative (11502-2)
8. Pathology Report Narrative (11526-1)

Chief Complaint:

Shortness of breath.

History of the Present Illness:

Mr. ■ is a previously healthy 56-year-old gentleman who presents with a four day history of shortness of breath, hemoptysis, and right-sided chest pain. He works as a truck driver, and the symptoms began four days prior to admission, while he was in Jackson, MS. He drove from Jackson to Abilene, TX, the day after the symptoms began, where worsening of his dyspnea and pain prompted him to go to the emergency room. There, he was diagnosed with pneumonia and placed on Levaquin 500 mg daily and Benzonatate 200 mg TID, which he has been taking for two days with only slight improvement. He then drove from Abilene back to Greensboro, where he resides, and continued to experience shortness of breath, right sided chest pain, and hemoptysis. He presented to an urgent care office in town today, and was subsequently transferred to the Moses Cone ER due to the provider’s suspicion of PE.

The right-sided pain is located midway down his ribcage, below the axilla. This pain is sharp, about 7/10 in severity, and worsens with movement and cough. Pressing on the chest does not recreate the pain. He feels that the pain has improved somewhat over the past two days. The hemoptysis has been unchanged since it began; there is not frank blood, but his sputum has been consistently blood-tinged. The blood seems redder at night. The dyspnea has been severe, and it is difficult for him to walk more than across a room. He states that he feels as though there is a “rattling” in his chest. At baseline, he experiences no dyspnea on exertion and has no history of COPD or other respiratory problem. He is a smoker, smoking a little less than a pack a day for thirty-five years. Past history is notable for the fact that he experienced transient left lower leg swelling – from below the knee down – and pain several weeks ago during a cross-country haul. He also notes a four day history of decreased appetite, poor sleep, and subjective fever and chills, with a measured fever of 103 in the hospital in Abilene. He had a bout of pneumonia about two months ago, but has been healthy for the most part and denies any chronic medical conditions. Currently he is fairly comfortable, with morphine helping with the pain. He has no history of a clotting disorder, no cardiac history, and denies any chest trauma or aspiration. He has had no sick contacts.

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Past Medical History:

1. Hernia repair
2. Bilateral thumb surgeries, secondary to two separate injuries sustained while working with machinery

Medications:

No regular medications, over-the-counter medications, or supplements. Has taken two days of the medications prescribed by the ER in Abilene: Levaquin 500 mg daily and Benzonatate 200 mg TID.

Review of Systems:

Constitutional: Denies changes in weight, fatigue, night-sweats.

HEENT: No changes in vision, nasal discharge, headache.

CV: No palpitations, left-sided chest pain/pressure, edema.

Resp: See HPI

GI: No nausea, vomiting, diarrhea, constipation.

GU: No dysuria, increased frequency.

Neuro: No weakness, confusion, numbness, dizziness.

MSK: No weakness, arthralgias, myalgias.

Heme: No easy bruising, easy bleeding.

Skin: No new lesions or rashes.

Endocrine: No polydipsia, polyuria, heat/cold sensitivity.

Social History:

Mr. [REDACTED] is divorced, lives in [REDACTED], and has been a truck driver for many years. He lives with his brother and his brother's wife. He has several children, including a daughter who also lives in the area. His family is very supportive and they get along well. As per HPI, he smokes a little less than a pack per day and has for thirty-five years. He does not drink alcohol or use other drugs. He tries to be physically active, and his favorite type of exercise is line dancing. He has “excellent” insurance through his employer.

Family History:

No family history of lung disease or clotting disorders.

Mother died at 92 of “old age.”

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Assessment & Plan:

Previously healthy 56 year old man presenting with shortness of breath, hemoptysis, and right-sided chest pain, with CT images demonstrating pulmonary emboli and infiltrates.

1. Shortness of breath: The symptoms of dyspnea, hemoptysis, and the right-sided chest pain could all be explained by a pulmonary embolus, and the CT shows evidence that confirms this diagnosis. Management should be to supply supplemental oxygen to maintain adequate oxygen saturation, to control Mr. ■'s chest pain, and to anticoagulate him. Anticoagulation with low molecular weight heparin and Coumadin could be started simultaneously, with both therapies should overlapping for at least five days; we will seek to obtain an INR between 2 and 3. The patient is currently hemodynamically stable, without evidence of right heart strain, so thrombolysis does not seem indicated at this time. With his history of long drives back and forth to California, Mr. ■ is at risk for venous stasis, which is a major source of DVT's; he even presents a history of lower leg pain and swelling. Testing for various causes of hypercoagulability would also be reasonable: these include deficiencies of proteins C and S, antithrombin III, lupus anticoagulant or anticardiolipin antibodies, and factor V mutations, prothrombin mutation, lipoprotein a, and hyperhomocystinemia. Another important factor to consider would be malignancy, especially as Mr. ■ is a long-time smoker. Screening for lung, colon, and prostate cancer would be appropriate.
2. Possible pneumonia: Mr. ■ has a history of being febrile, and was diagnosed with pneumonia in Abilene (obtaining these records will be helpful), but does not currently have an elevated white count. The CT demonstrates an infiltrate, possibly consistent with pneumonia. Sputum cultures and blood cultures would be reasonable to obtain, though since he has been taking an antibiotic for presumed community acquired pneumonia, this may be low-yield. His vitals should be monitored regularly. It is possible that all pulmonary symptoms stem from the pulmonary embolus, but it would be reasonable to continue the antibiotic therapy that he has already started in order to cover the possibility of an infectious component.
3. Anemia: Admission labs show hemoglobin at 11.5, with MCV of 89.5, so this anemia is normocytic. The patient denies any history of anemia when questioned.

Notes are a rich source of info

Research and Applications

Real world evidence in cardiovascular medicine: ensuring data validity in electronic health record-based studies

Tina Hernandez-Boussard,^{1,2,3} Keri L Monda,^{4,5} Blai Coll Crespo,⁴ and Dan Riskin^{1,3,6}

Table 1. Cohort identification of diseases and procedures stratified by EHR-S and EHR-U data^a

Cohort	Occurrence		EHR-S			EHR-U		
	Concept	Patient	Recall (%)	Precision (%)	F1-score (%)	Recall (%)	Precision (%)	F1-score (%)
Hyperlipidemia	2471	837	65.2	99.3	78.7	98.2	99.4	98.8
Hypercholesterolemia	1899	478	55.1	98.0	70.5	90.4	98.8	94.4
Coronary artery disease	1427	465	67.5	99.4	80.4	94.6	96.2	95.4
Diabetes mellitus	4502	1377	80.6	97.9	88.4	97.0	92.6	94.8
Myocardial infarction	523	282	29.8	86.2	44.2	90.4	76.5	82.9
Chronic kidney disease	640	101	40.8	97.6	57.6	92.9	97.9	95.3
Stroke	693	307	36.5	97.2	53.0	95.7	79.6	87.0
Dementia	317	103	62.1	100.0	76.6	93.1	90.0	91.5
Cataract	240	85	28.6	100.0	44.4	96.1	94.9	95.5
CABG ^b	194	73	32.2	100.0	48.7	96.6	95.0	95.8

^aAll comparisons were significant at $P < .0001$.

^bCoronary artery bypass graft.

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Imaging

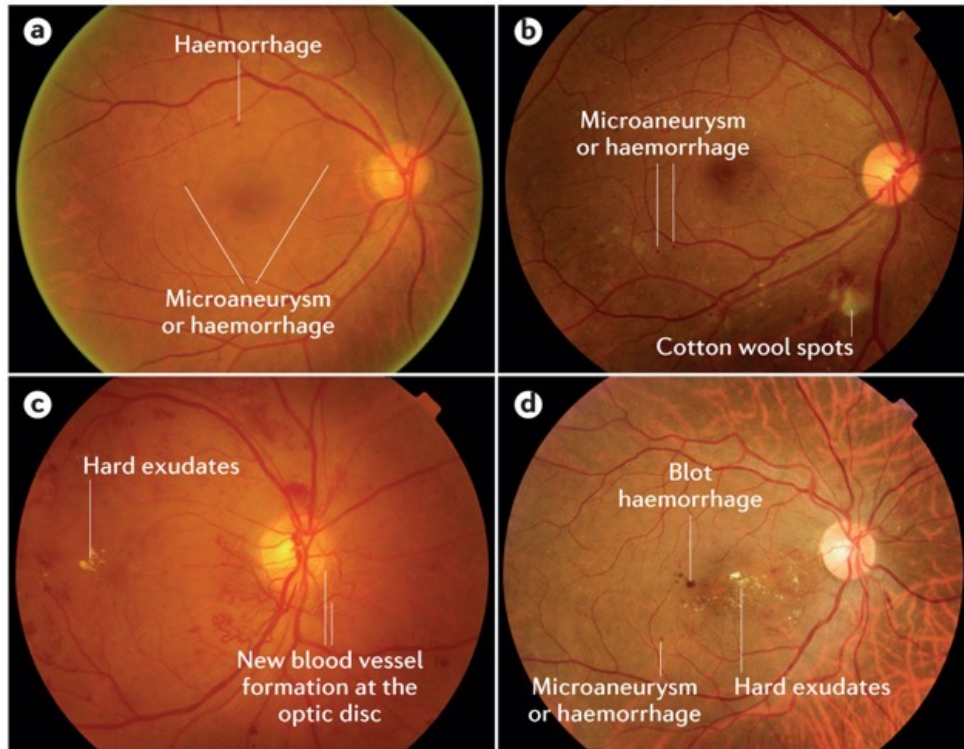
- A CT is worth a thousand words
- Radiologist's interpretation (e.g. radiology report):
 - Distills known, salient info
 - But can miss relevant signal



Large non-occlusive pulmonary embolus draped over the bifurcation of the main pulmonary artery; a saddle pulmonary embolus.

<https://radiopaedia.org/cases/pulmonary-embolism-saddle-1?lang=us>

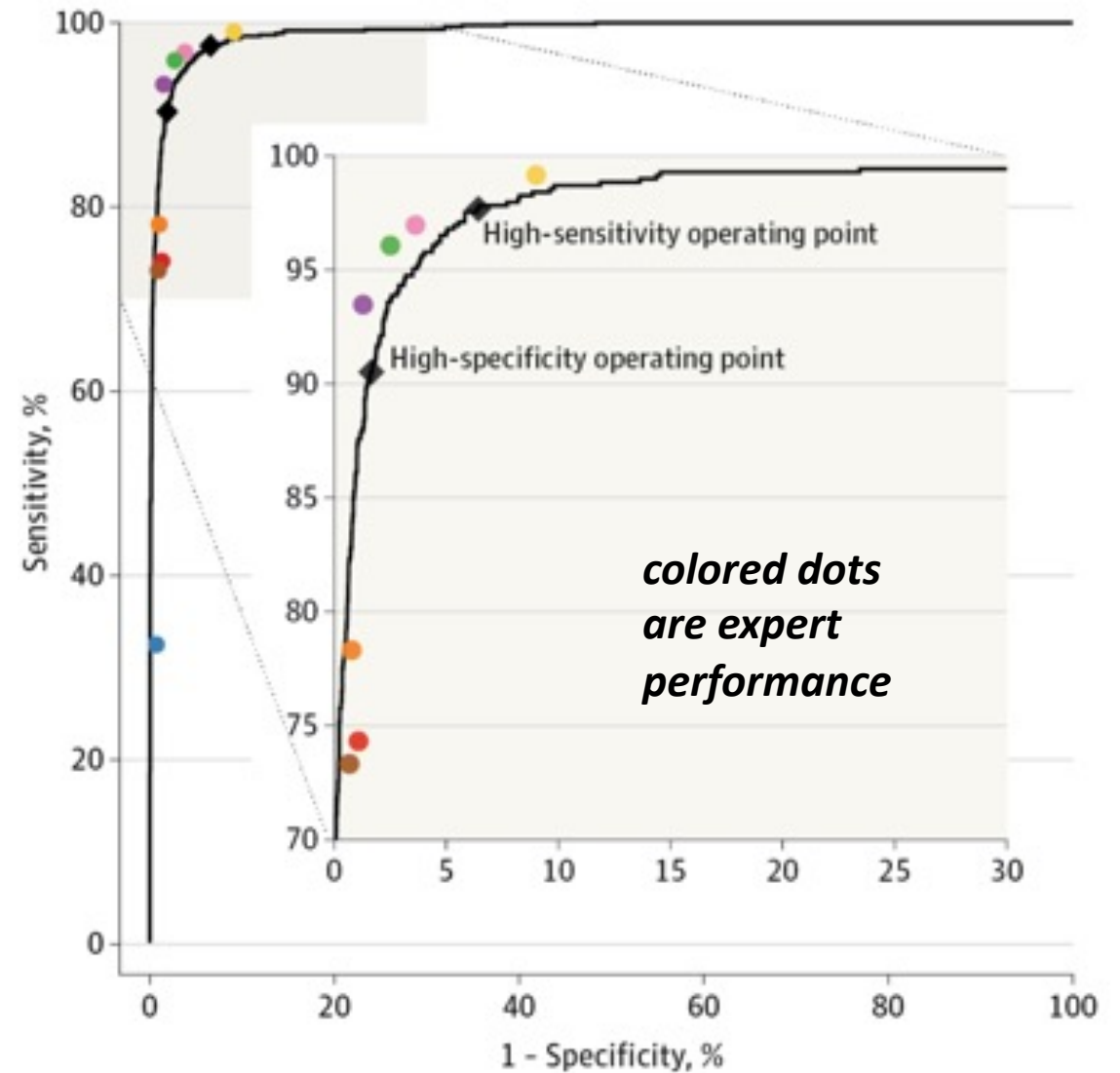
Computer vision vs clinical experts



Nature Reviews | Disease Primers

Wong, T. Y. *et al.* (2016) Diabetic retinopathy
Nat. Rev. Dis. Primers doi:10.1038/nrdp.2016.12

A EyePACS-1: AUC, 99.1%; 95% CI, 98.8%-99.3%

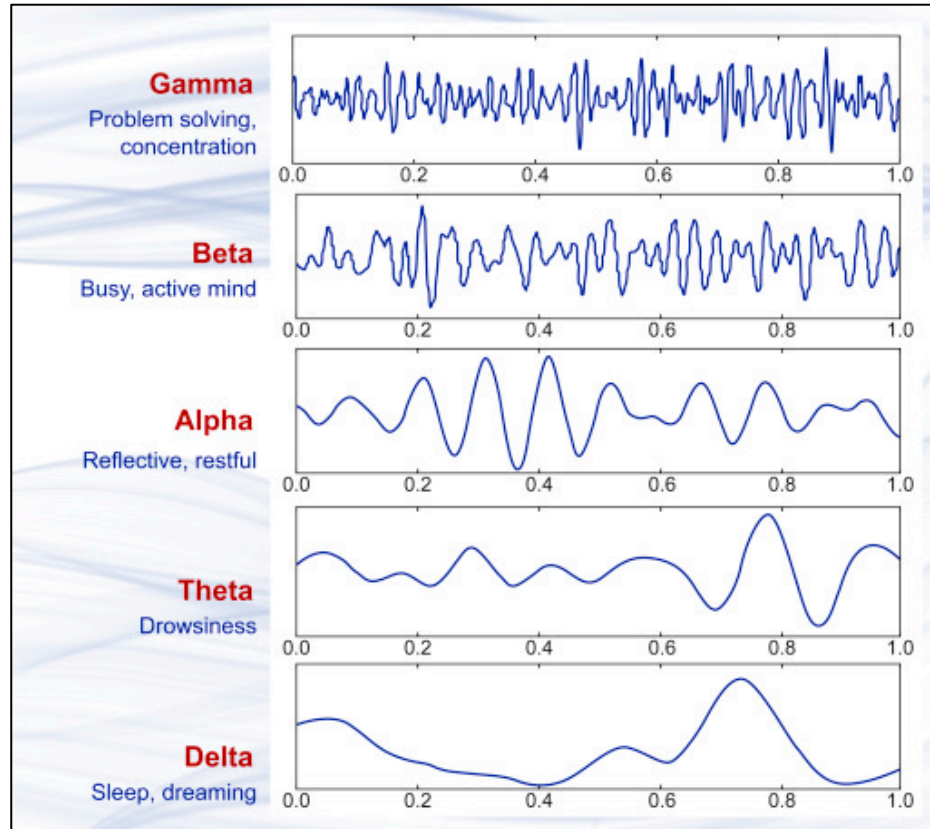


Gulshan V, Peng L, Coram M, et al. Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs. *JAMA*. 2016;316(22):2402-2410. doi:10.1001/jama.2016.17216

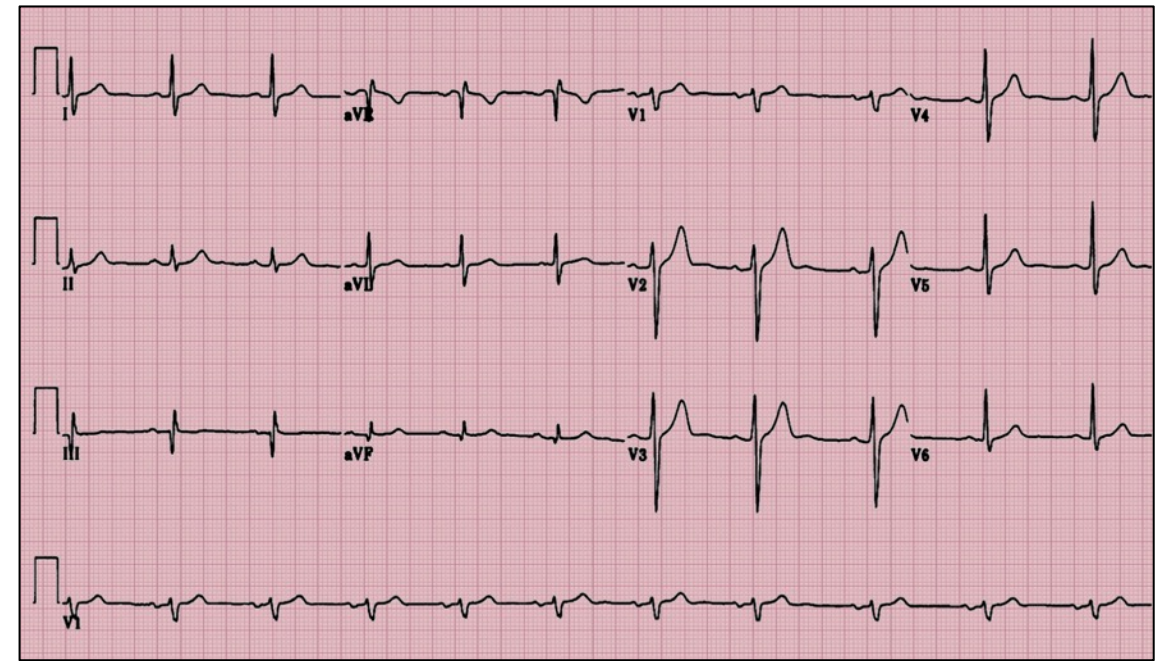
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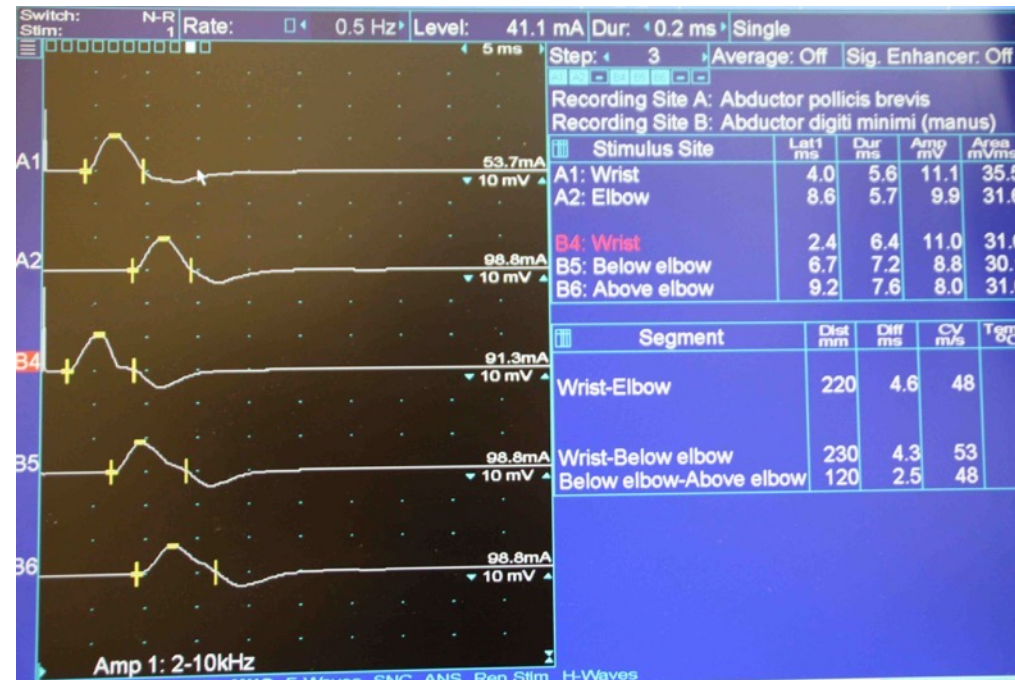
Electrophysiology



EEG



EKG



EMG

Why use unstructured data?

- Reason 1: it contains information not found in structured fields
 - clinical/domain knowledge helps gauge whether/when this info will be useful
 - *example*: use radiological images when predicting COPD prognosis
- Reason 2: it can be more accessible, no expert interpretation needed
 - *example*: predict malignancy from smartphone images of skin lesions
- Reason 3: it is unaffected (or less affected) by provider decision-making
 - *example*: predict need for surgery from imaging of small bowel obstruction

Why might we need many modalities?



Imaging

PCOR_RSH_DIAGNOSIS_VW EXAMPLE															
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Structured EHR data

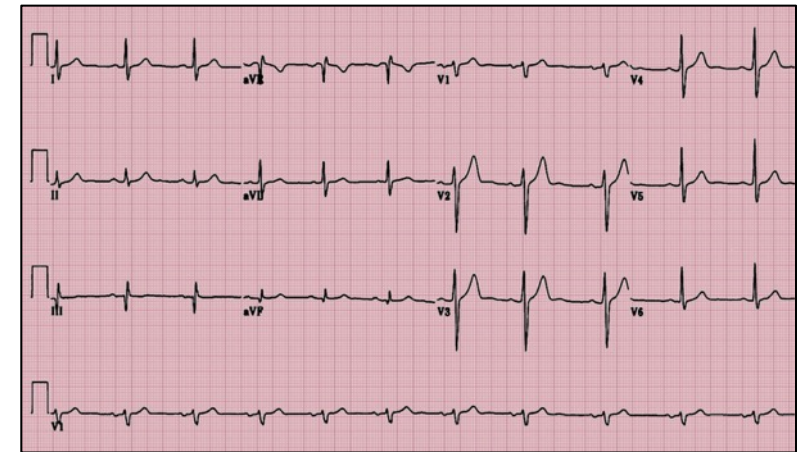
Clinical notes

Chief Complaint:

Shortness of breath.

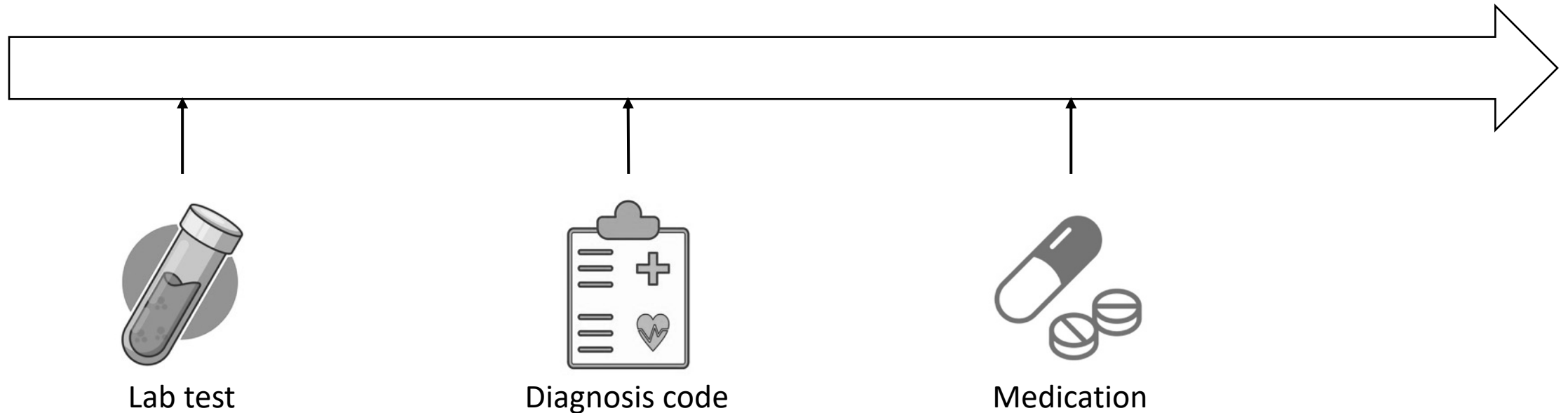
History of the Present Illness:

Mr. ■ is a previously healthy 56-year-old gentleman who presents with a four day history of shortness of breath, hemoptysis, and right-sided chest pain. He works as a truck driver, and the symptoms began four days prior to admission, while he was in Jackson, MS. He drove from Jackson to Abilene, TX, the day after the symptoms began, where worsening of his dyspnea and pain prompted him to go to the emergency room. There, he was diagnosed with pneumonia and placed on Levaquin 500 mg daily and



EKG and other physiologic time-series

A patient timeline of structured data



A patient timeline of multi-modal data

Imaging



Note

Chief Complaint:
Dyspnea at rest.

History of the Present Illness:
A 67-year-old male, previously healthy, 60-year-old gentleman who presents with a four-day history of dyspnea at rest, hemoptysis, and right-sided chest pain. He works as a truck driver, and the symptoms began the day prior to admission, while he was in Jackson, MS. He drove from Jackson to Atlanta, GA, the day after the symptoms began, where worsening of the dyspnea and pain prompted him to go to the emergency room. There, he was diagnosed with pneumonia and placed on Levofloxacin 500 mg daily and Ibuprofen 200 mg TID, which he has been taking for two days with only slight improvement. He then drove from Atlanta back to Chattanooga, where he resides, and continued to experience shortness of breath, right-sided chest pain, and hemoptysis. He presented to an urgent care office in town today, and was subsequently transferred to the Johns Voss ER due to the provider's suspicion of PE.

The right-sided pain is located midway down the thorax, below the axilla. This pain is sharp, about 7/10 in severity, and worsens with movement and cough. Pressing on the chest does not relieve the pain. He feels that the pain has improved somewhat over the past two days. The hemoptysis has been unchanged since it began; there is not frank blood, but he expects he has been consistently blood-tinged. The blood seems reddish at night. The dyspnea has been worse, and it is difficult for him to walk more than across a room. He states that he feels as though there is a "weight" in his chest. At baseline, he experiences no dyspnea on exertion and has no history of COPD or other respiratory problems. He is a smoker, smoking a half pack a day for thirty-five years.

Past history is notable for the fact that he experienced constant full lower-leg swelling—first below the knee down—and pain several weeks ago during a cross-country haul. He also has a first-degree burn on his right forearm, pain, and mild depressive mood and skills, with a measured level of 10 in the hospital in Atlanta. He had a brief of pneumonia about two months ago. He has been healthy for the most part and denies any chronic medical conditions. Currently he is fairly comfortable, with morphine helping with his pain. He has no history of a chronic disease, no cardiac history, and denies any chest trauma or aspiration. He has had no risk contacts.

Past Medical History:
1. Hypertension
2. History of asthma
3. History of chronic sinusitis, secondary to two separate injuries sustained while working with machinery

Medications:
No regular medications, over-the-counter medications, or supplements. Has taken two days of the medications prescribed by the ER in Atlanta: Levofloxacin 500 mg daily and Ibuprofen 200 mg TID.

Electrophysiology



Lab test



Diagnosis code



Medication

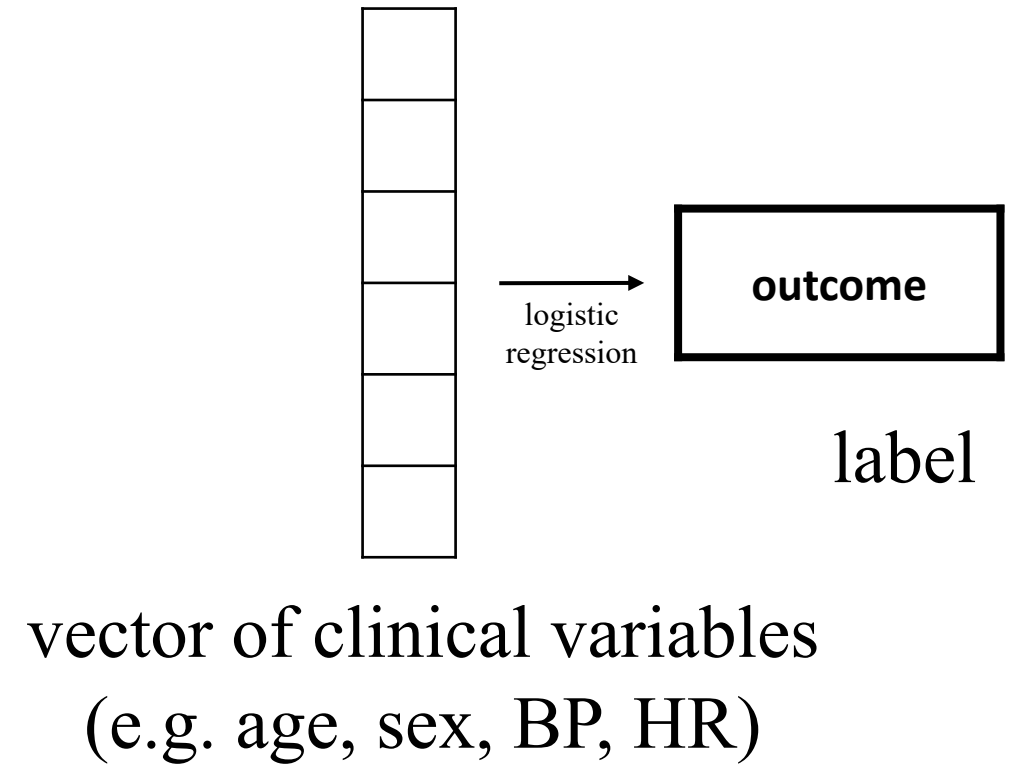
Task-specific, single-modality AI

Image processing

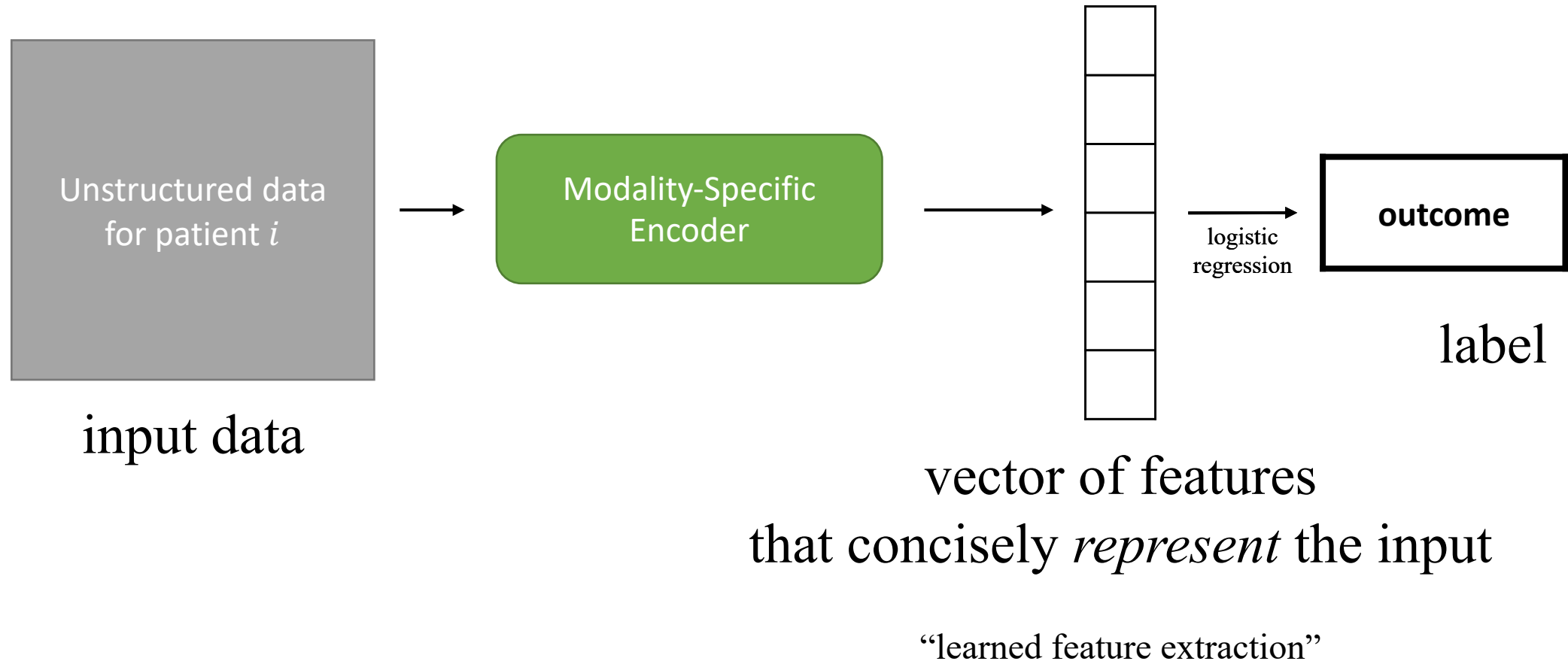
Natural language processing

Structured data processing

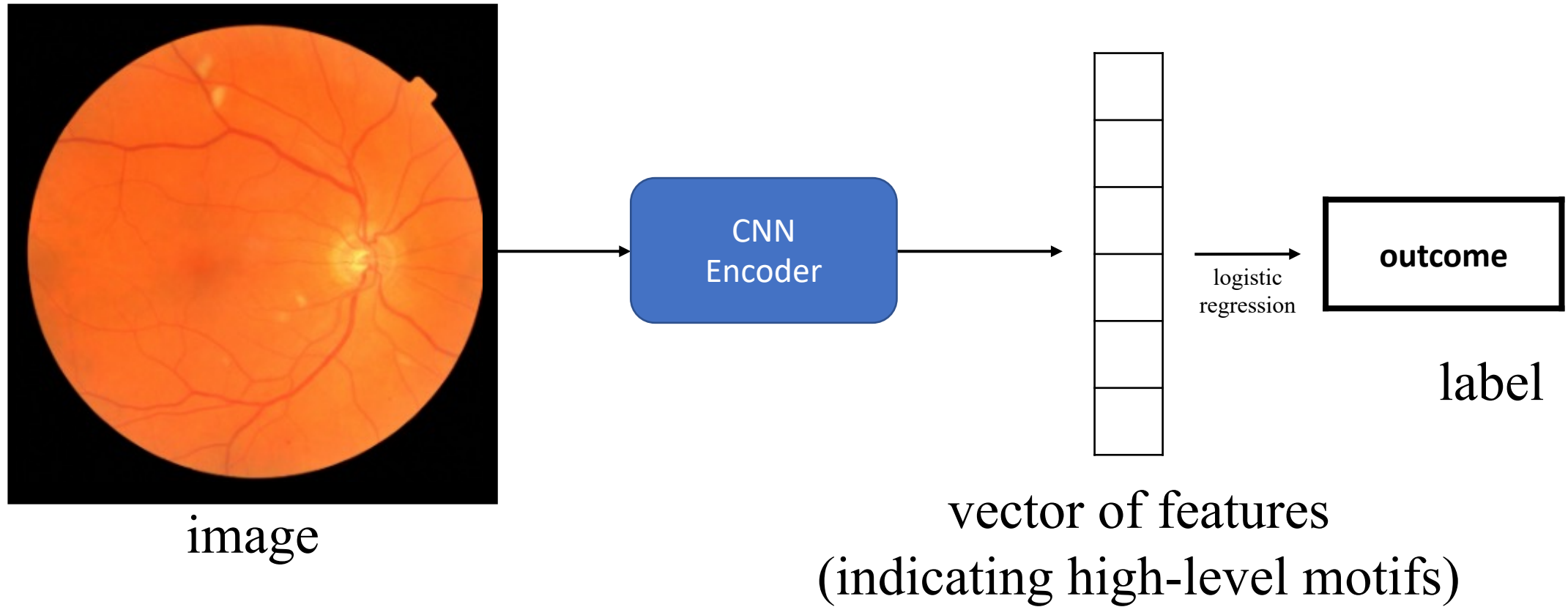
A typical prediction model



Learning *representations* of complex medical data



Learning *representations* of images



Learning *representations* of clinical notes

Chief Complaint:
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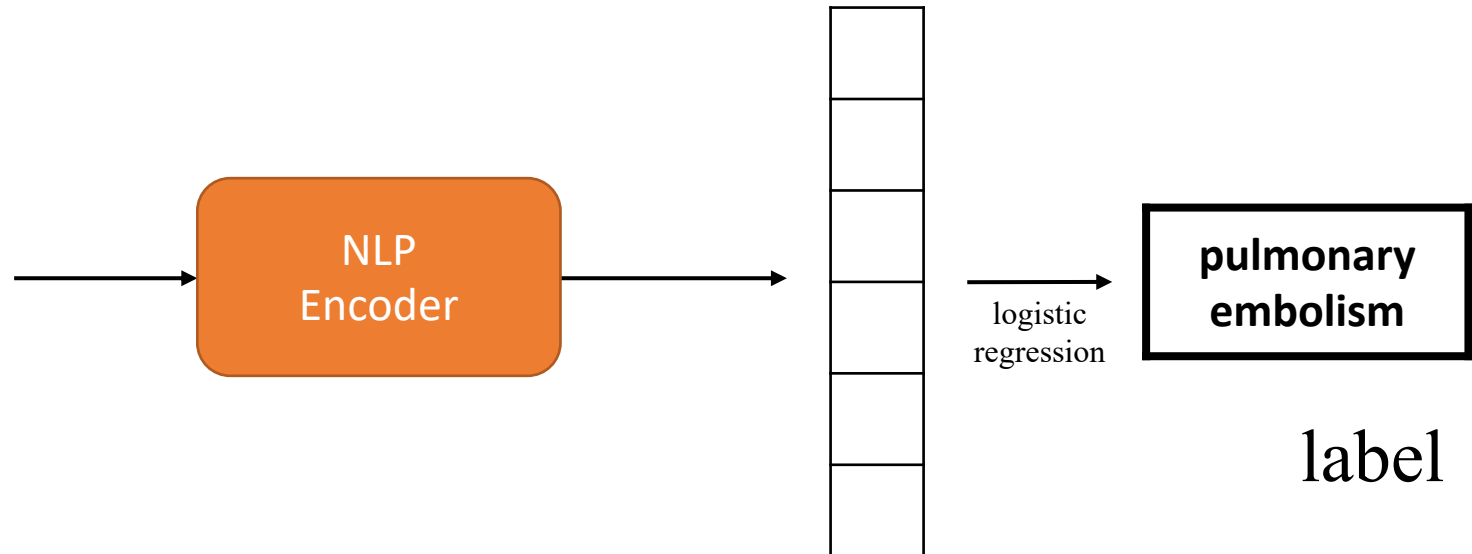
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Past Medical History:

1. Hernia repair
2. Bilateral thumb surgeries, secondary to two separate injuries sustained while working with machinery

Medications:
No regular medications, over-the-counter medications, or supplements. Has taken two days of the medications prescribed by the ER in Abilene: Levaquin 500 mg daily and Benzonatate 200 mg TID.

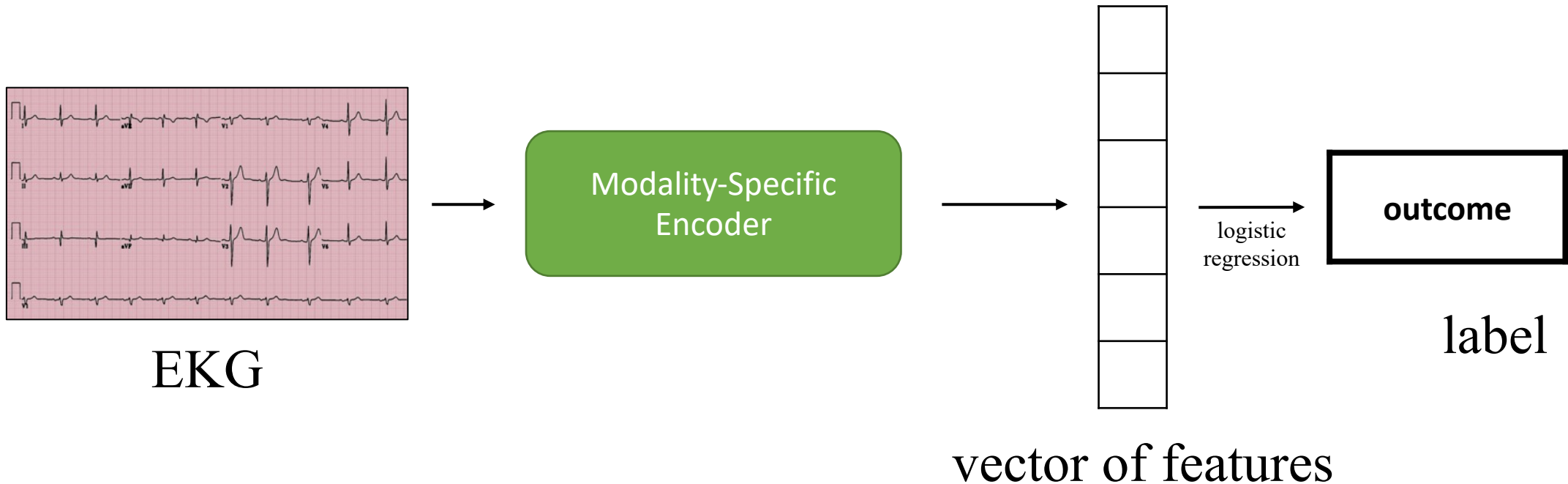
document



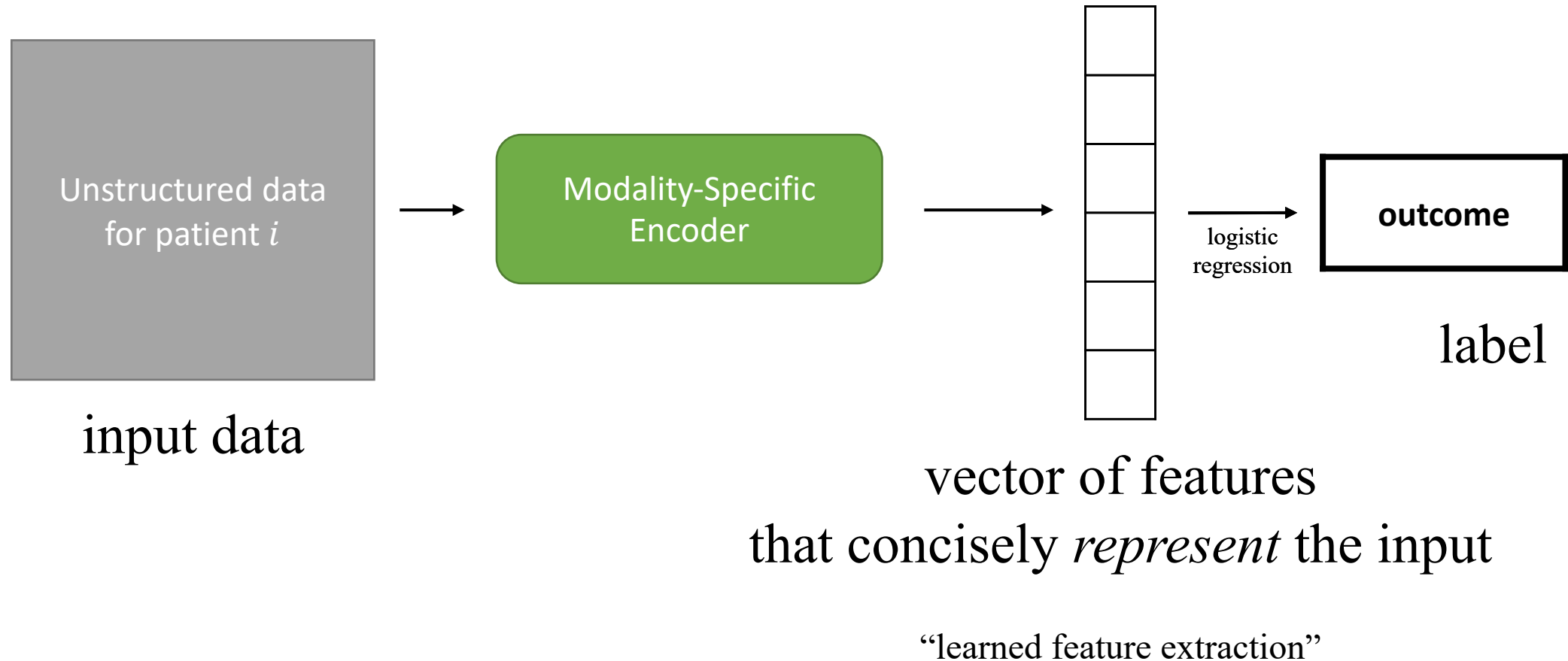
vector of features

(old: indicating word counts)
(new: indicating semantic features)

Learning *representations* of other modalities

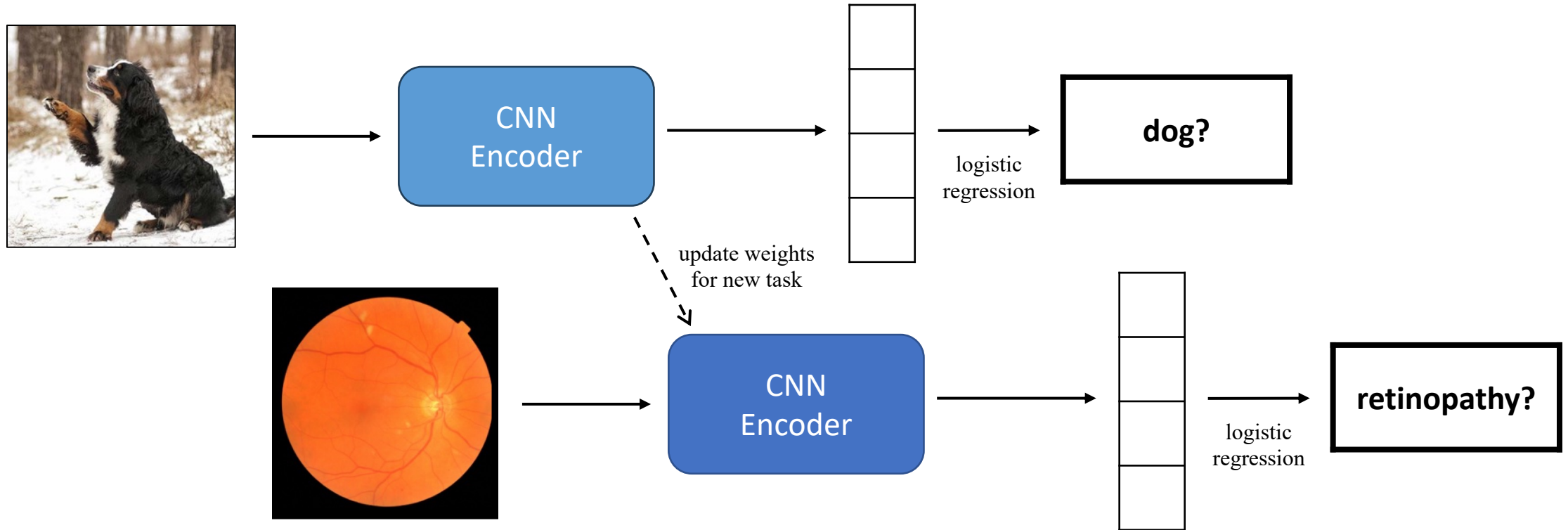


Learning *representations* of complex medical data



Transfer learning:

Train on one task, fine-tune on another



The model remains task-specific, but can be adapted to new tasks ad hoc
This tends to be helpful, because some features are important to many tasks

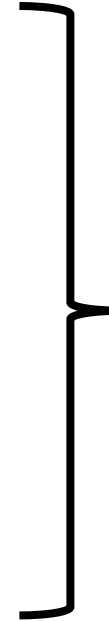
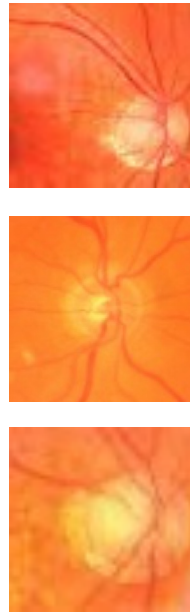
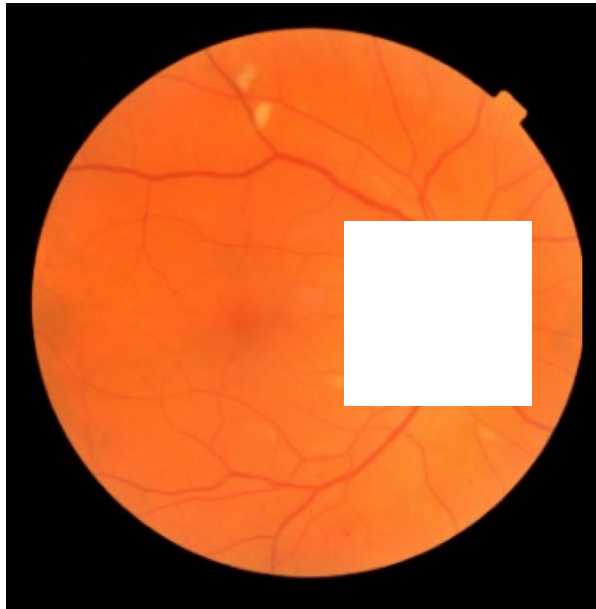
Task-agnostic, single-modality AI

Image processing

Natural language processing

Structured data processing

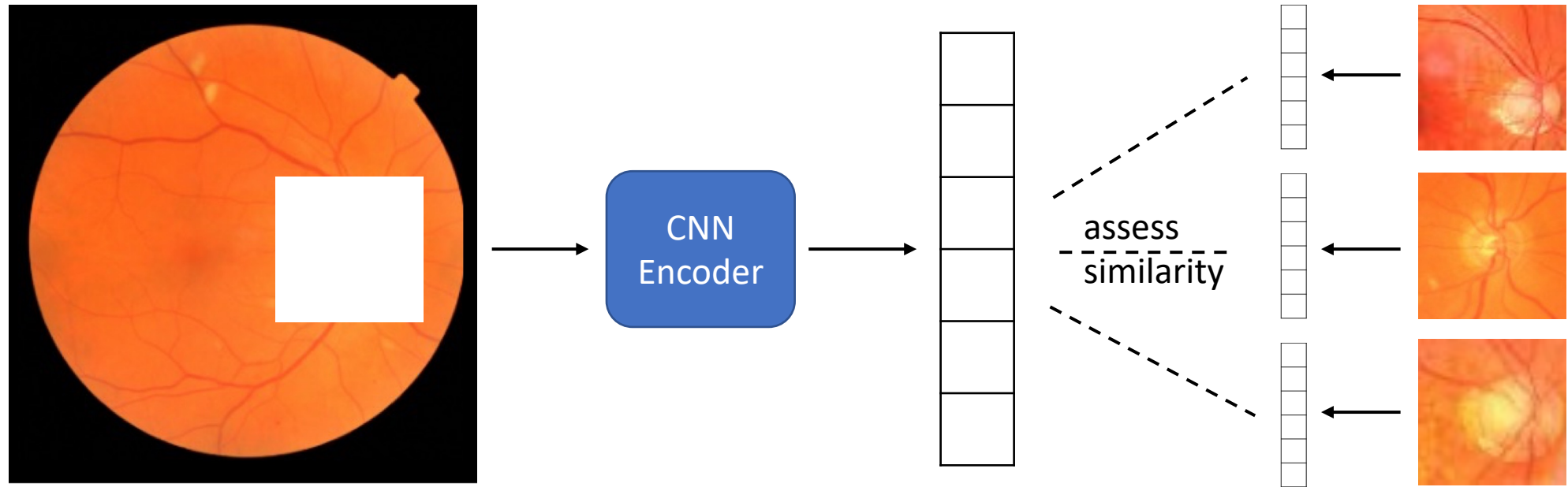
Self-supervision: predict the missing *patch*



Which is correct?

Filling in the blank – or identifying the correct completion – requires an understanding of fundoscopic images

Self-supervision: predict the missing *patch*



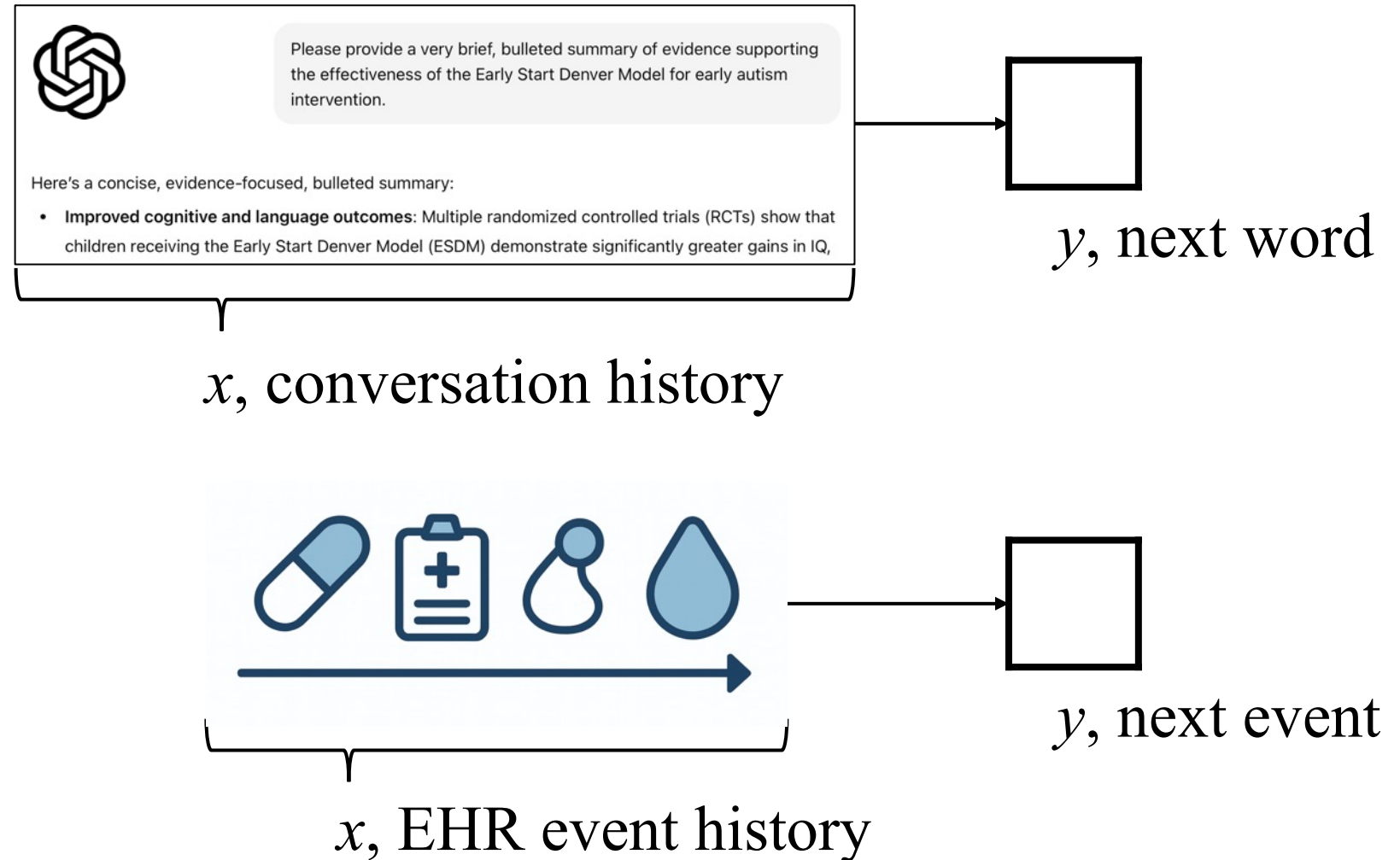
Filling in the blank – or identifying the correct completion – requires an understanding of fundoscopic images *helpful for many prediction tasks*

Self-supervision: predict the next *word* or *event*

EHR Foundation models

Intuition:

Learn representations sufficient to predict the next (or missing) EHR event; these can then be adapted for specific prediction tasks.



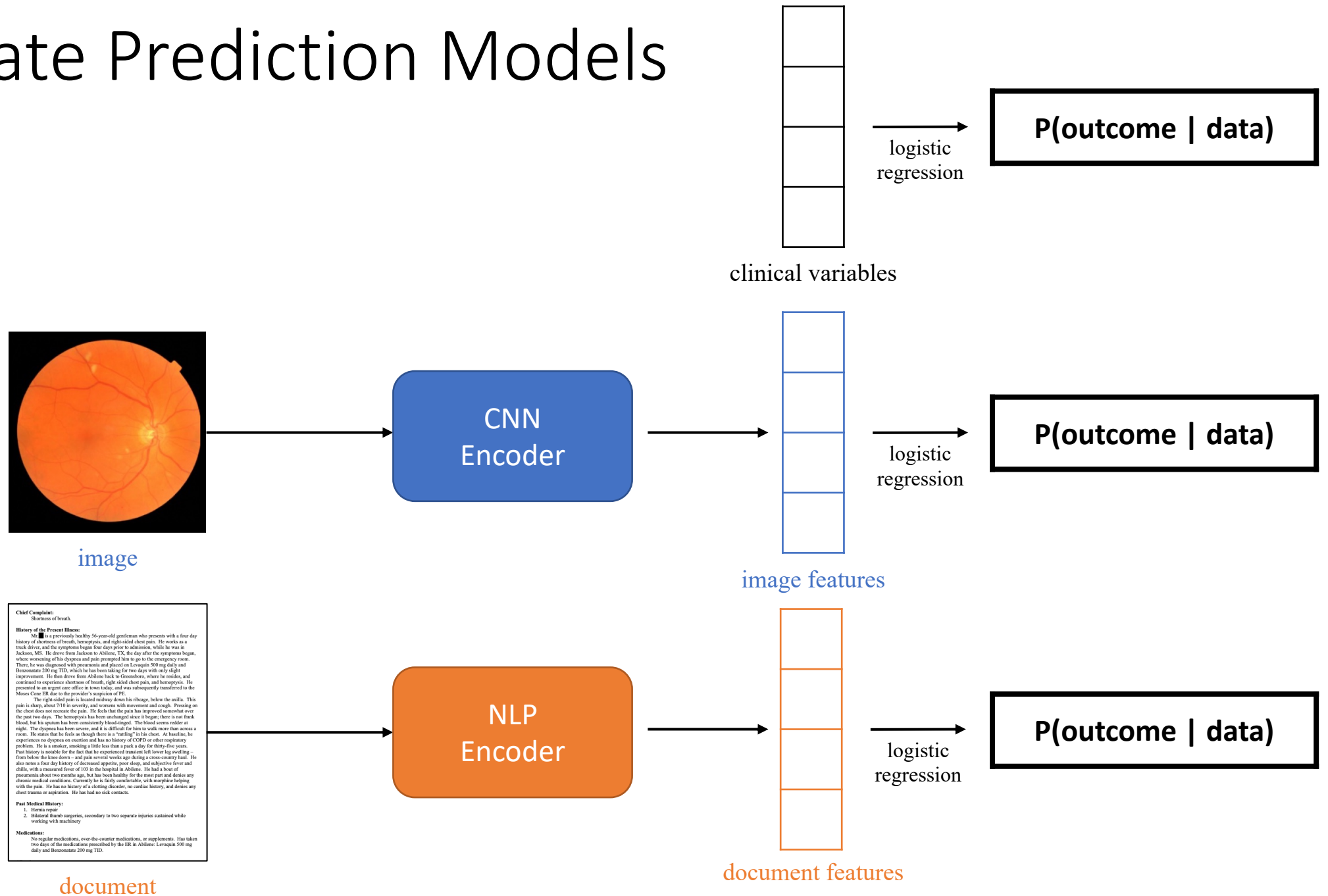
Task-specific, multi-modality AI

Image processing

Natural language processing

Structured data processing

Separate Prediction Models



Late fusion



image

an old gentleman who presents with a four day
and right-sided chest pain. He works as a
days prior to admission, while he was in
bilious. TX, the day after the symptoms began,
prompted him to go to the emergency room.
and placed on Levaquin 500 mg daily and
en taking for two days with only slight
back to Groningen, where he resides, and
a, right sided chest pain, and hemoptysis. He
today, and was subsequently transferred to the
town of PE.
Jaw down his ribcage, below the axilla. This
resolves with movement and cough. Pressing on
is that the pain has improved somewhat over
in unchanged since it began, there is not flank
blood-tinged. The blood seems redder at
is difficult for him to walk more than across a
there is a "rattling" in his chest. At baseline, he
is no history of COPD or other respiratory
less than a pack a day for thirty-five years.
experienced transient left lower leg swelling ...
not weeks ago during a cross-country haul. He
appetite, poor sleep, and subjective fever and
hospital in Ahlme. He had a bout of
been healthy for the most part and denies any
fairly comfortable, with morphine helping
ing disorder, no cardiac history, and denies any
risk contacts.

y to two separate injuries sustained while

counter medications, or supplements. Has taken
bed by the ER in Ahlme: Levaquin 500 mg

document

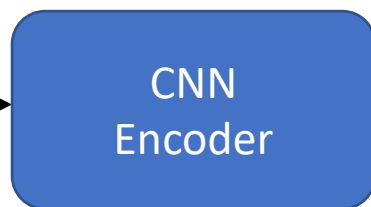


image features

logistic
regression

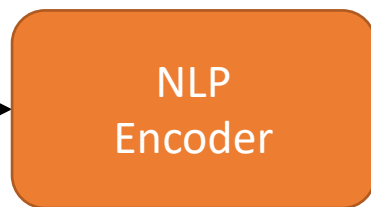


clinical variables

logistic
regression



aggregate predictions

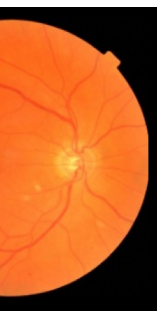


document features

logistic
regression



Late fusion



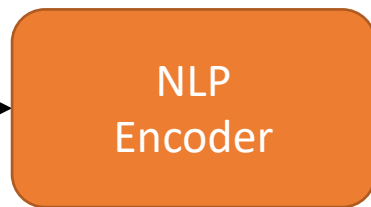
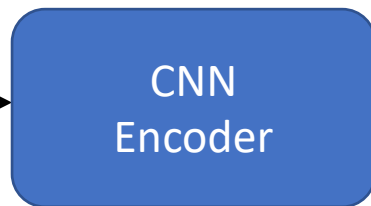
image

our old gentleman who presents with a four day
and right-sided chest pain. He works as a
days prior to admission, while he was in
bilious. 72, the day after the symptoms began,
prompted him to go to the emergency room.
and placed on Levaquin 500 mg daily and
en taking for two days with only slight
back to Greenboro, where he resides, and
a, right sided chest pain, and hemoptysis. He
today, and was subsequently transferred to the
town of PE.
Jaw down his ribcage, below the axilla. This
resolves with movement and cough. Pressing on
is that the pain has improved somewhat over
no unchanged since it began; there is not flank
blood-tinged. The blood seems redder at
is difficult for him to walk more than across a
there is a "rattling" in his chest. At baseline, he
is no history of COPD or other respiratory
less than a pack a day for thirty-five years.
experienced transient left lower leg swelling ...
not weeks ago during a cross-country haul. He
appetite, poor sleep, and subjective fever and
hospital in Atlanta. He had a bout of
been healthy for the most part and denies any
fairly comfortable, with morphine helping
ing disorder, no cardiac history, and denies any
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y to two separate injuries sustained while

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bed by the ER in Atlanta: Levaquin 500 mg

document



clinical variables

image features

document features



logistic
regression

logistic
regression

logistic
regression

$P(\text{outcome} \mid \text{data})$

$P(\text{outcome} \mid \text{data})$

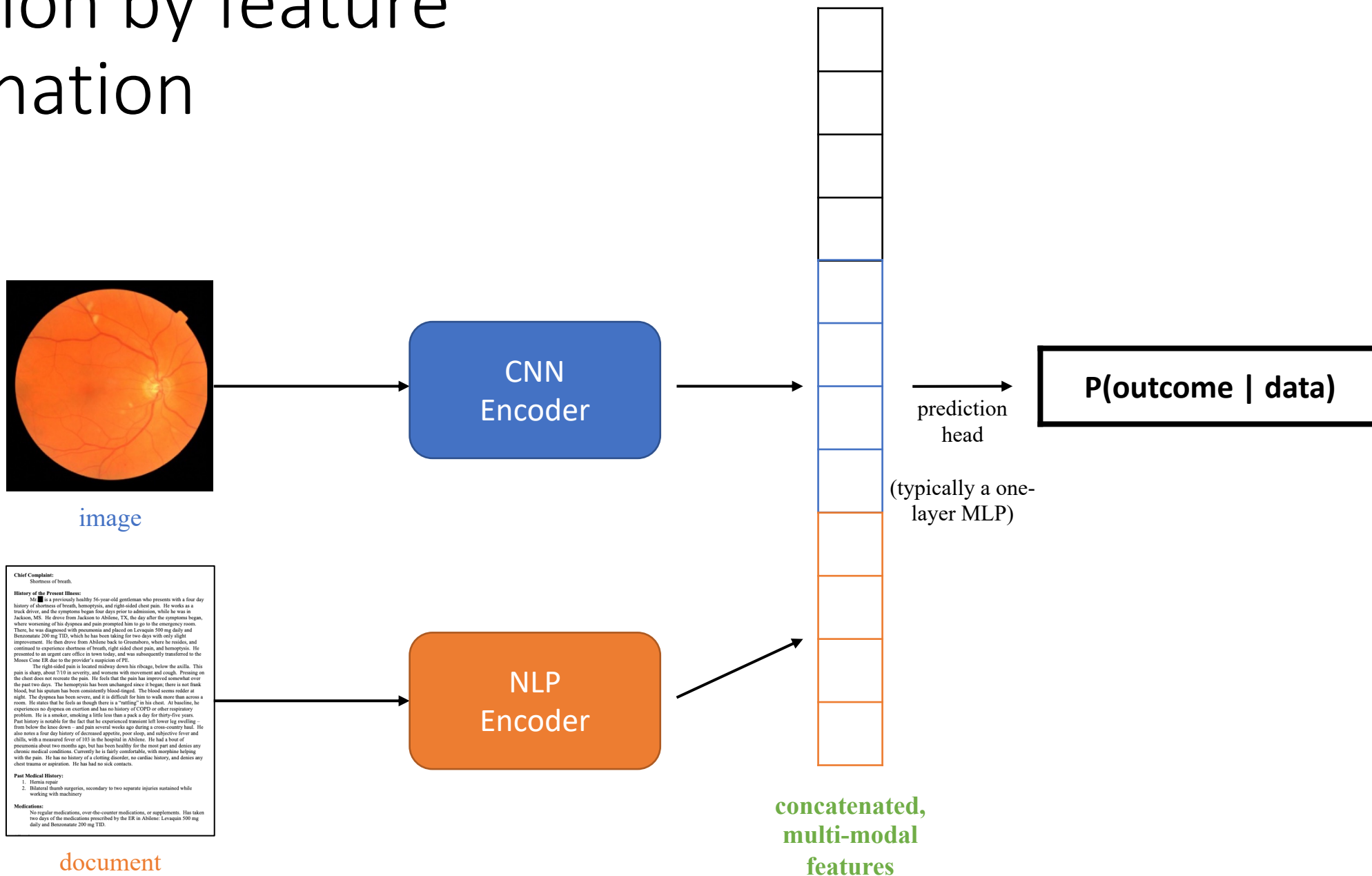
$P(\text{outcome} \mid \text{data})$

aggregate predictions

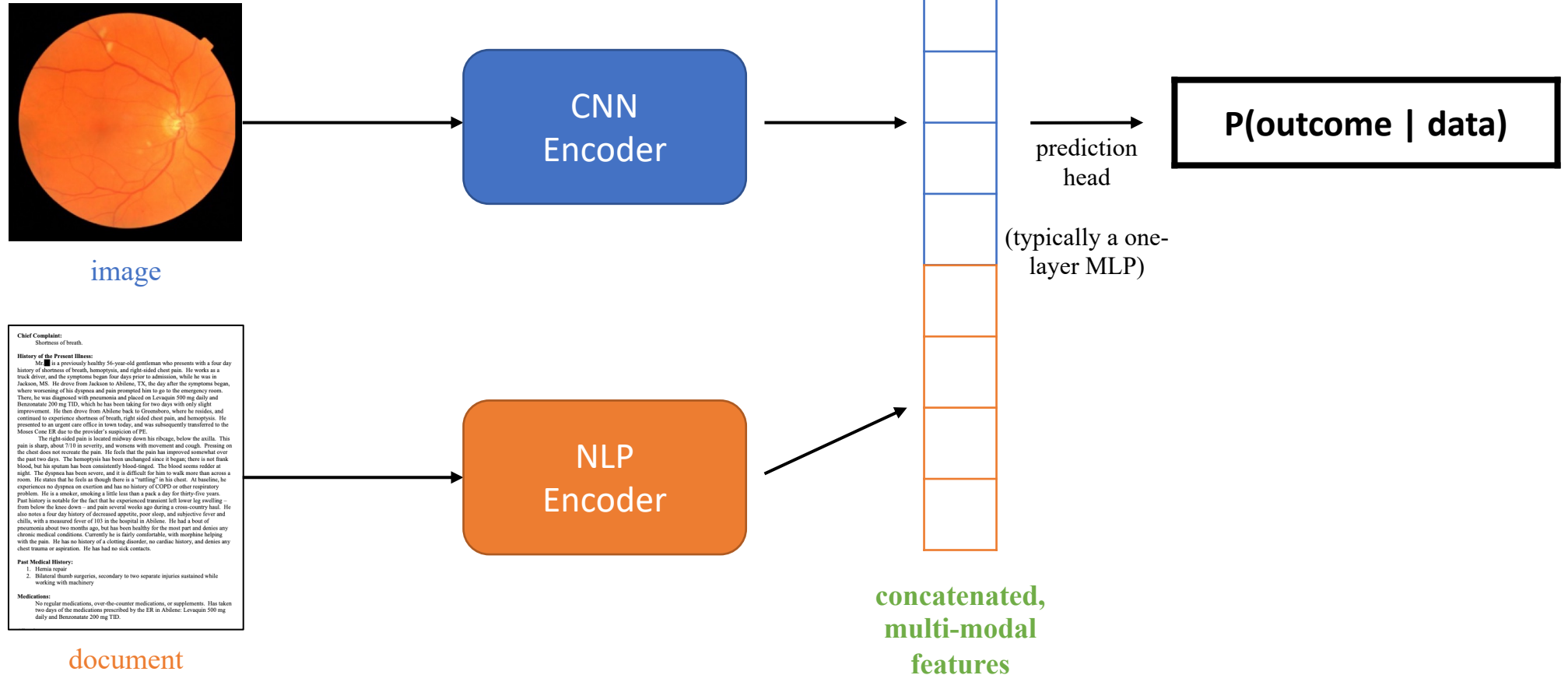
- Average
- Max (e.g., MIL)
- Meta-predictor (this is called *stacking*)
- Bayesian fusion

(a range of strategies,
some more complicated
than others, can be used
to aggregate)

Joint fusion by feature concatenation

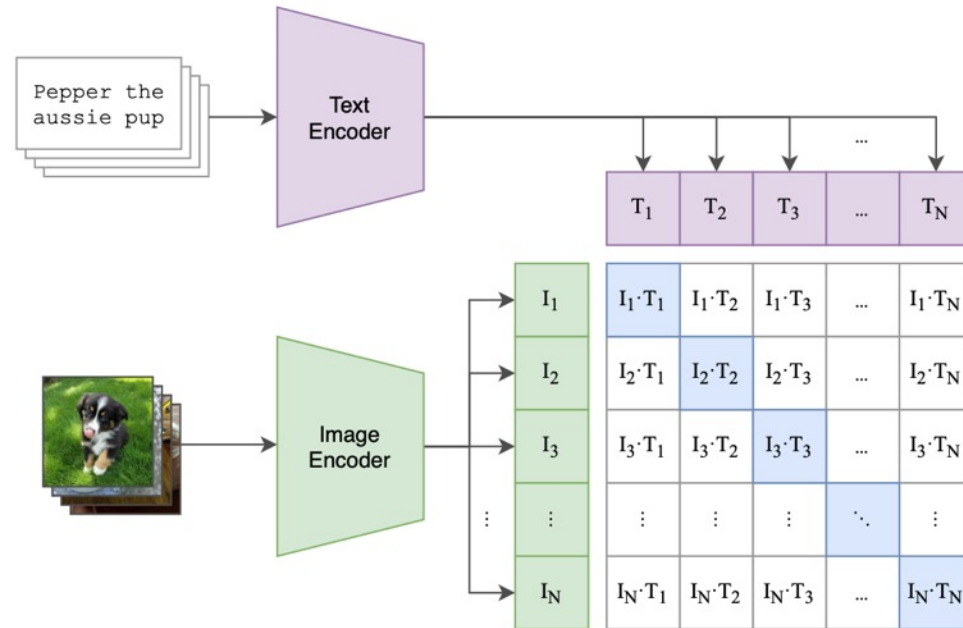


Joint fusion by feature concatenation

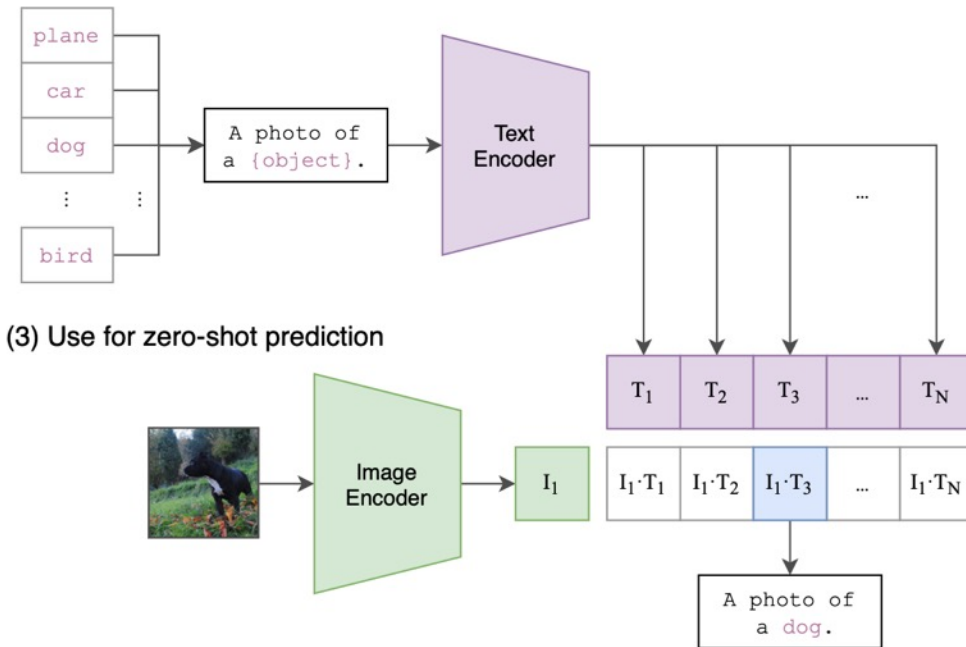


Joint fusion via contrastive learning

(1) Contrastive pre-training



(2) Create dataset classifier from label text

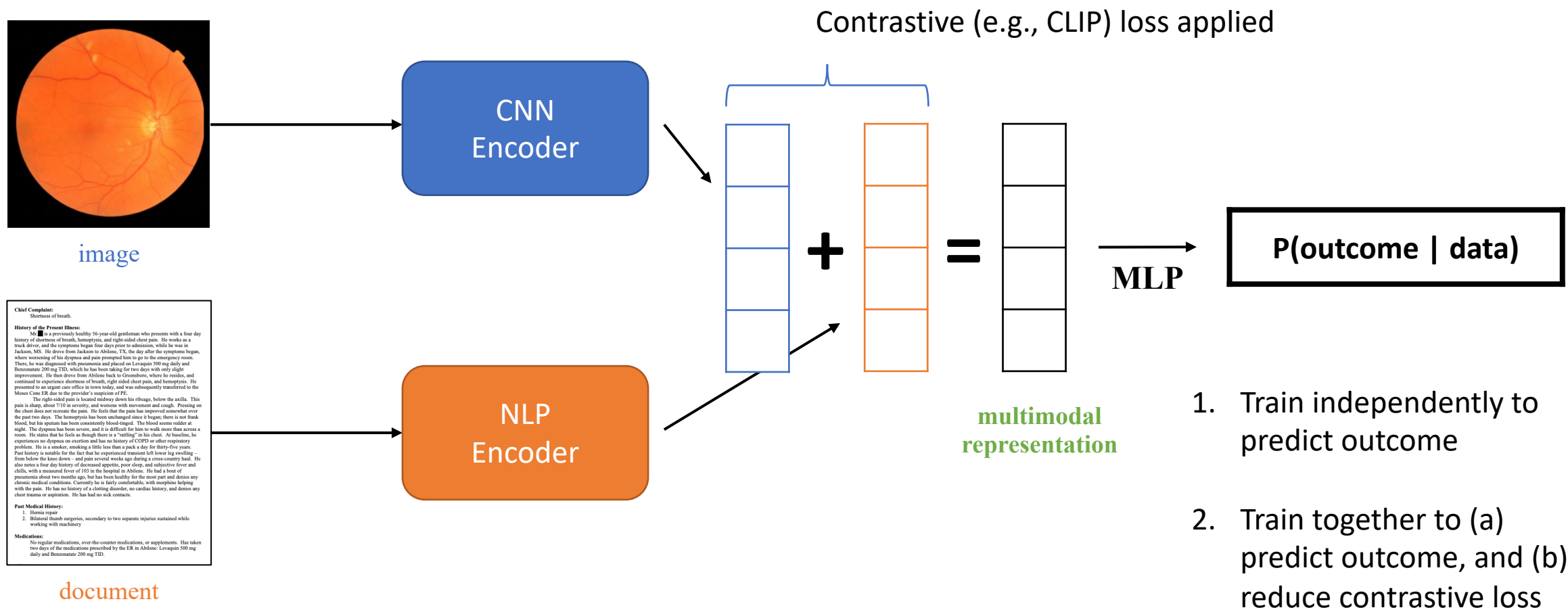


(3) Use for zero-shot prediction

Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes.

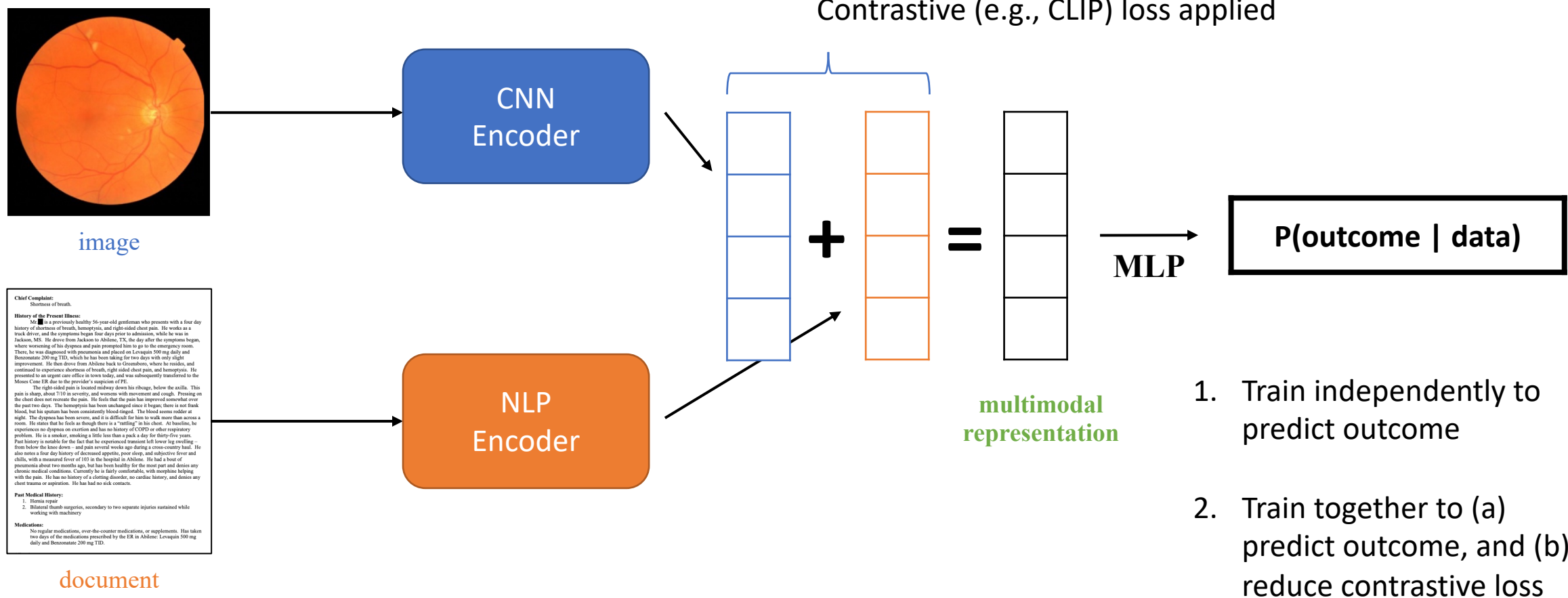
Radford A, Kim JW, Hallacy C, Ramesh A, Goh G, Agarwal S, Sastry G, Askell A, Mishkin P, Clark J, Krueger G. Learning transferable visual models from natural language supervision. In International conference on machine learning 2021 Jul 1 (pp. 8748-8763). PmLR.

Joint fusion in shared representation space



Joint fusion in shared representation space

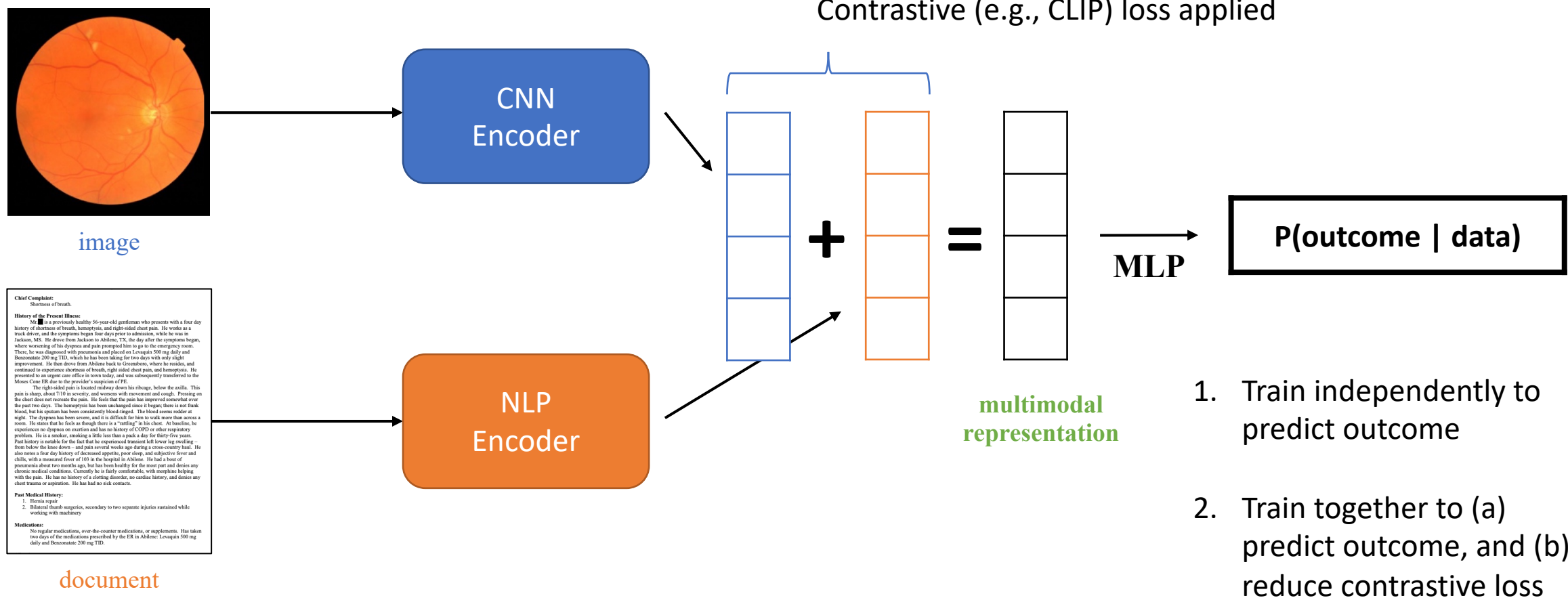
Q: What happens if we train to minimize the contrastive loss only?



Joint fusion in shared representation space

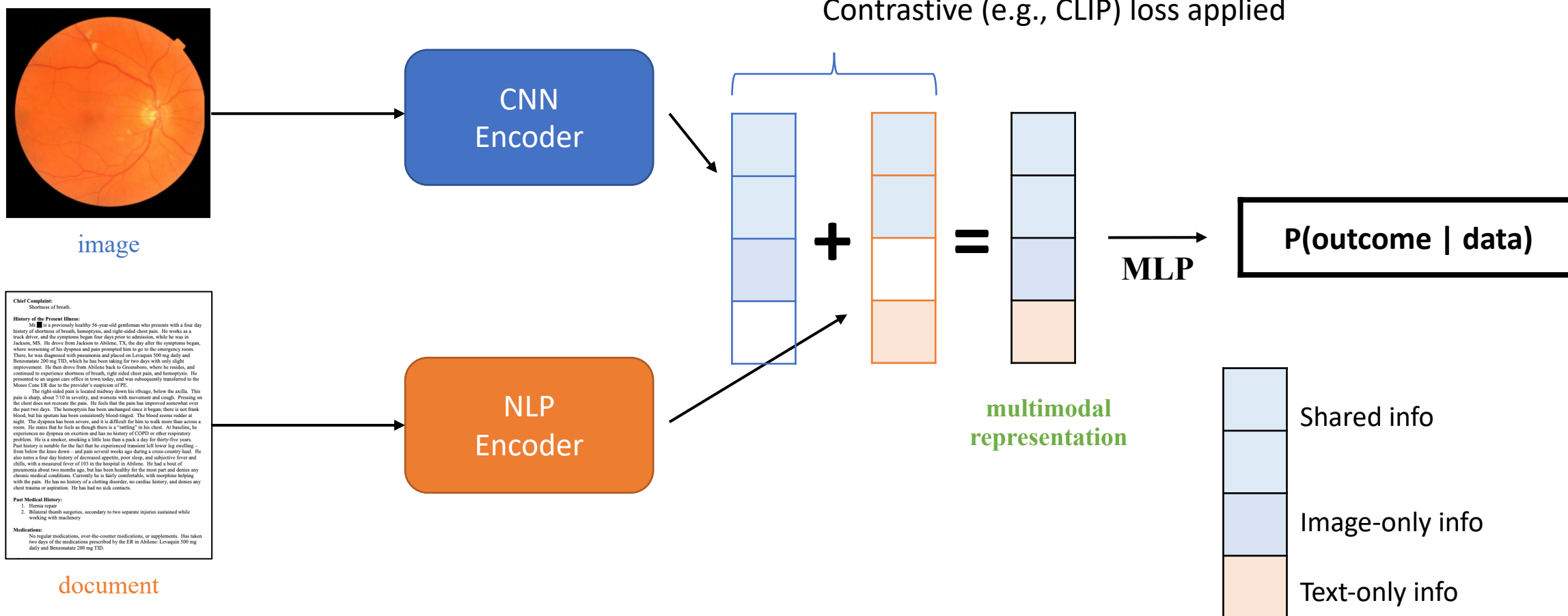
Q: What happens if we train to minimize the contrastive loss only?

A: we will lose *distinct* information between modalities that is predictive of the outcome



Joint fusion in shared representation space

1. Contrastive loss encourages shared info between modalities to concentrate in a subset of channels
2. Prediction loss prevents distinct info from being lost



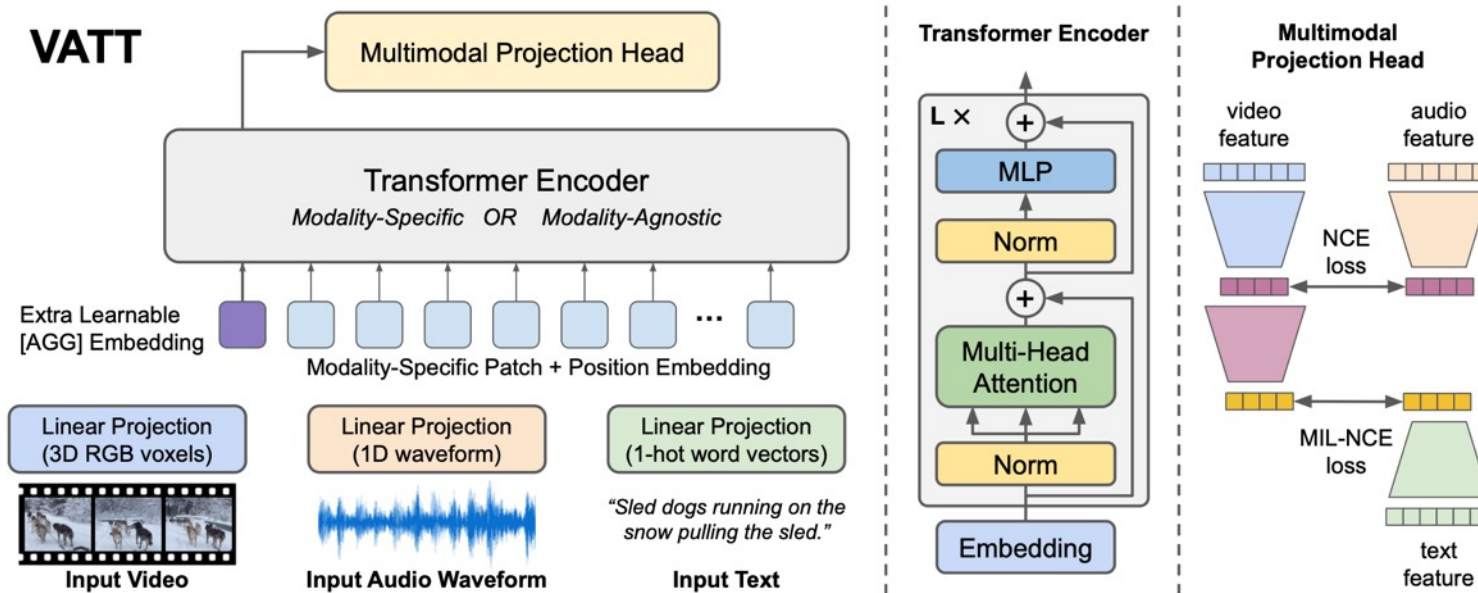
Task-agnostic, multi-modality AI

Image processing

Natural language processing

Structured data processing

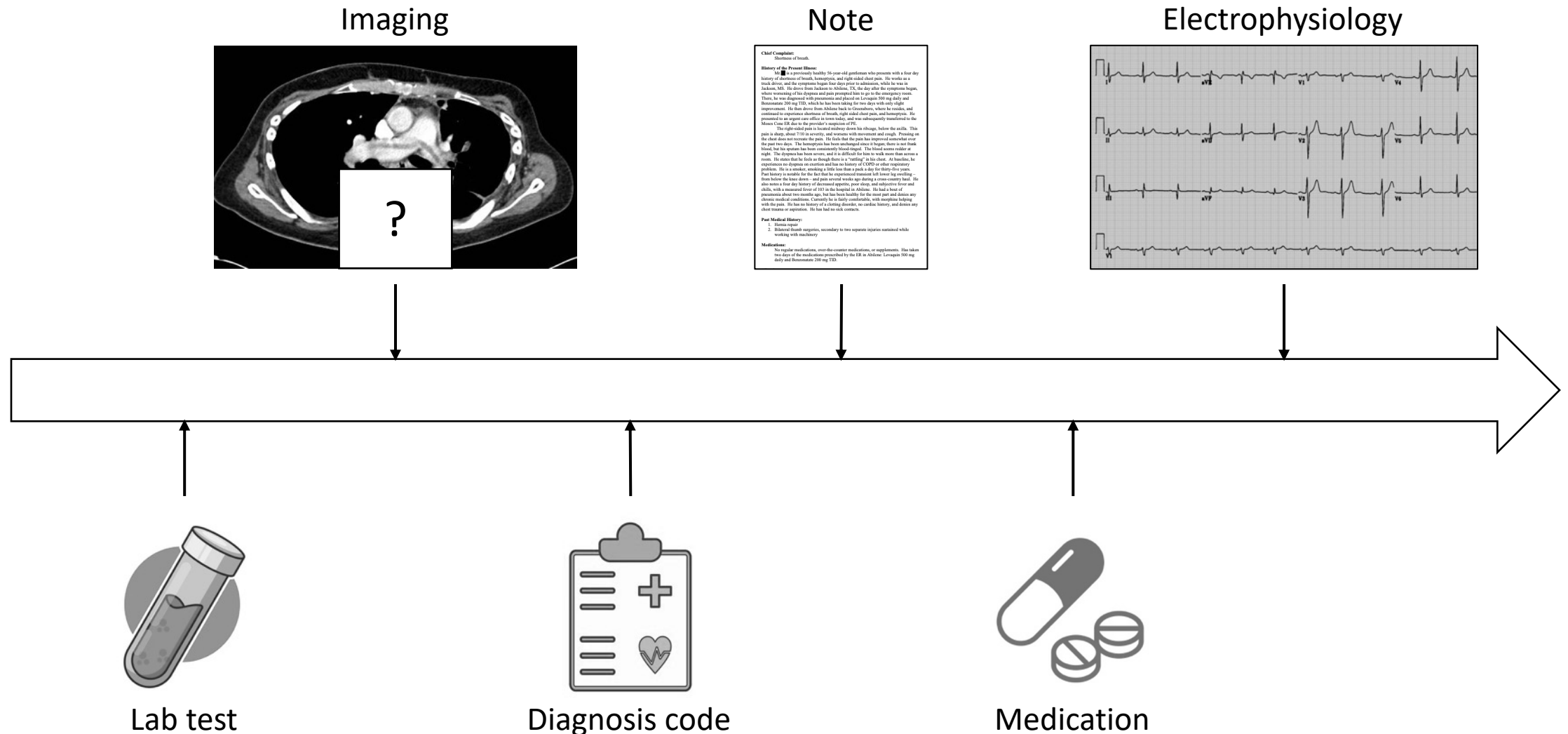
Frontier models have moved to early fusion



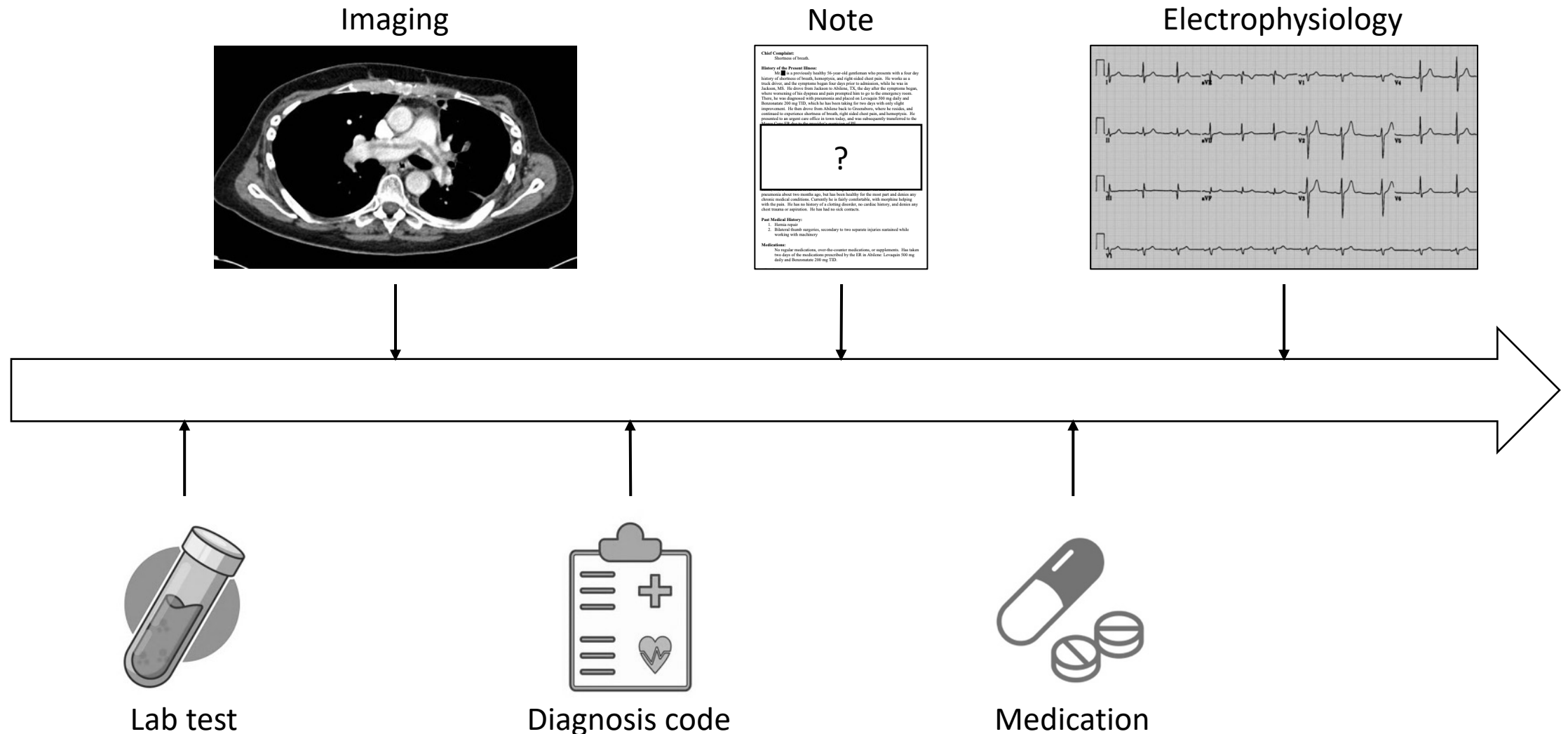
- Distinct linear projections map raw pixels and waveforms to a shared input embedding space
- Contrastive loss is applied to encourage output representations to be similar across modalities

Akbari, H., Yuan, L., Qian, R., Chuang, W.-H., Chang, S.-F., Cui, Y., & Gong, B. (2021). VATT: Transformers for Multimodal Self-Supervised Learning from Raw Video, Audio and Text. *Advances in Neural Information Processing Systems (NeurIPS)* 34.

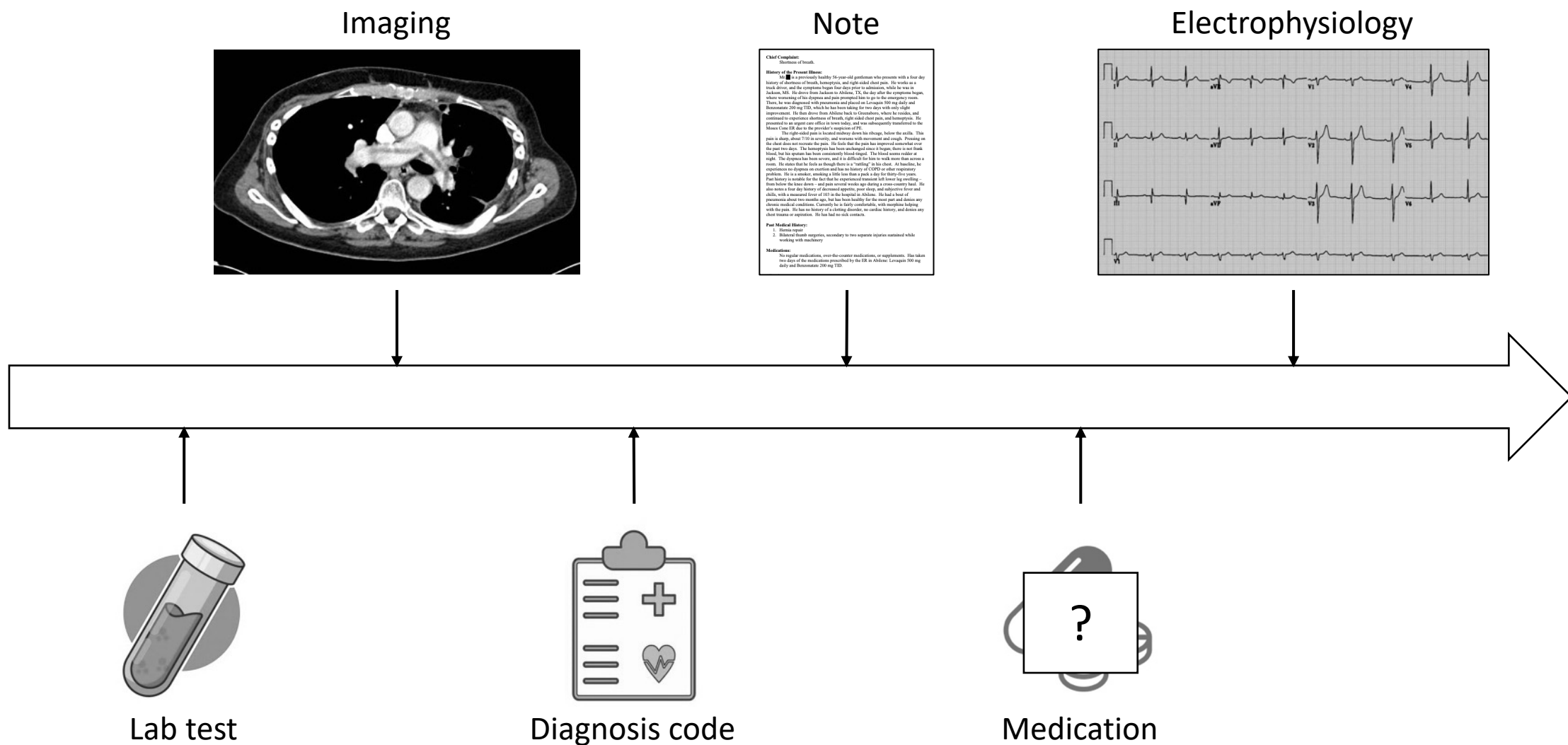
Self-supervision: predict the missing *multi-modal token*



Self-supervision: predict the missing *multi-modal token*



Self-supervision: predict the missing *multi-modal token*



Summary

- EHR and other medical data are comprised of multiple data types, or *modalities*
- Distinct modalities *complement one another* to enhance prediction
- We have moved from task-specific, uni-modal models to task-agnostic, multi-modal models that *understand health data* and support a wide range of analyses
- These approaches are highly data intensive up front, but can then be applied in small samples to improve prediction or interpretation

Thank you!

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